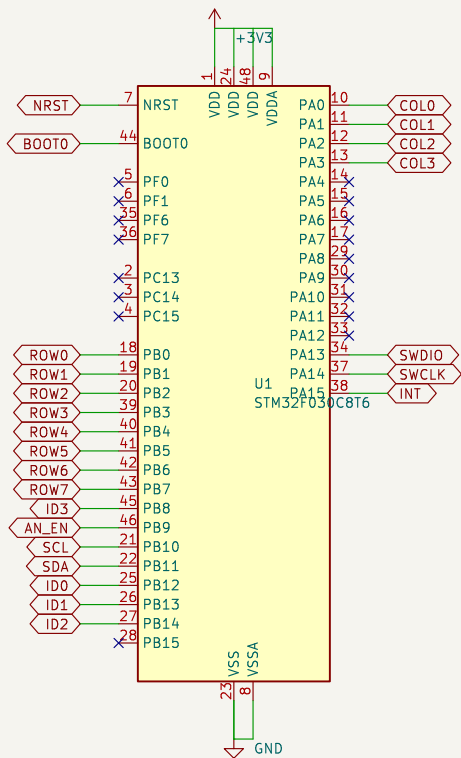
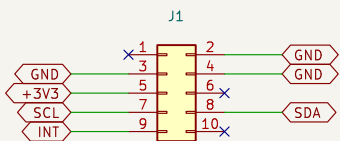


MCU – STM32F030C8T6 running the RetroPad firmware (I2C slave 0x6D)

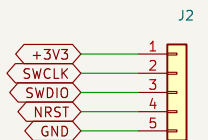


Tab5 Ext.Port1 (M5Stack Tab5 Keyboard P1 pinout) – verify pin 1 orientation before fab



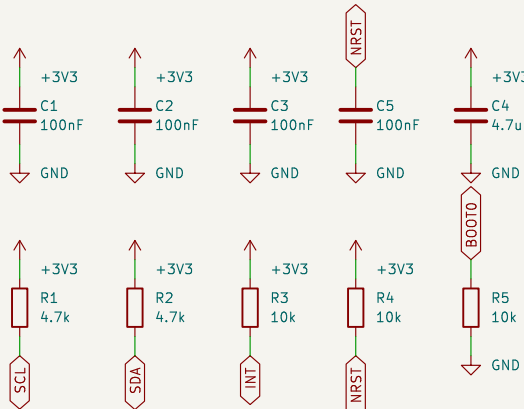
Tab5 Ext.Port1

SWD (same order as the keyboard's P2)

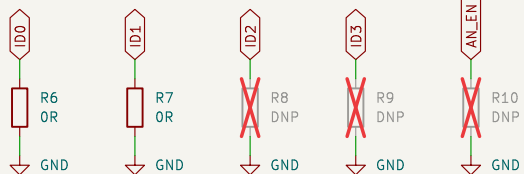


SWD

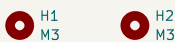
Decoupling, pull-ups, reset, BOOT0 (copied from the M5Stack keyboard)



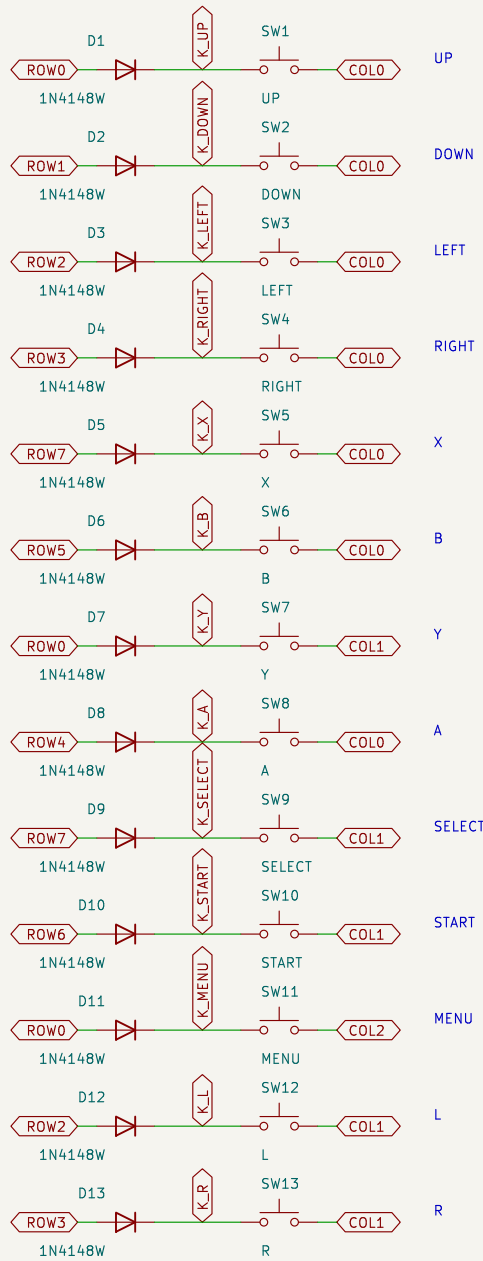
Console ID straps: OR to GND = bit set. SNES = ID 3 (fit R6, R7; R8, R9, R10 DNP)



Mounting holes (M3, 96 mm apart)



Button matrix – row driven high, column read with pull-down; diode anode on the row



Tab5 keyboard–port board: STM32F030 + RetroPad firmware, I2C 0x6D  
Generated by kicad\_gen/gen\_sch.py from gen\_pcb.py and layout.py

**RetroESP32–P4**

Sheet: /  
File: retropad\_snes.kicad\_sch

**Title: RetroPad SNES controller (console ID 3)**

Size: A3 Date: 2026–10–05  
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Rev: 0.1  
Id: 1/1